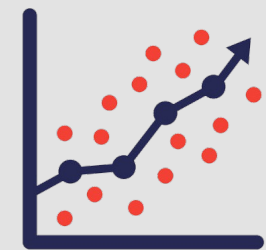


วิเคราะห์ข้อมูลด้วย RapidMiner

Asst.Prof. Dr.Damrongrit Setsirichok

Faculty of Digital Technology

Chitralada Technology Institute



Dataset
(ชุดข้อมูล)

Sample

Attribute/ Feature

Label

Outlook	Temperature	Humidity	Windy	Play
sunny	85	85	FALSE	no
sunny	80	90	TRUE	no
overcast	83	78	FALSE	yes
rain	70	96	FALSE	yes
rain	68	80	FALSE	yes
rain	65	70	TRUE	no
overcast	64	65	TRUE	yes
sunny	72	95	FALSE	no
sunny	69	70	FALSE	yes
rain	75	80	FALSE	yes
sunny	75	70	TRUE	yes
overcast	72	90	TRUE	yes
overcast	81	75	FALSE	yes
rain	71	80	TRUE	no

Example

Attribute Type

Attribute Type		Description	Examples
Categorical (Qualitative)	Nominal	The values of a nominal attribute are just different names; i.e., nominal values provide only enough information to distinguish one object from another. (=, $\neq$)	zip codes, employee ID numbers, eye color, gender
	Ordinal	The values of an ordinal attribute provide enough information to order objects. ($<$, $>$)	hardness of minerals, { <i>good</i> , <i>better</i> , <i>best</i> }, grades, street numbers
Numeric (Quantitative)	Interval	For interval attributes, the differences between values are meaningful, i.e., a unit of measurement exists. (+, -)	calendar dates, temperature in Celsius or Fahrenheit
	Ratio	For ratio variables, both differences and ratios are meaningful. (*, /)	temperature in Kelvin, monetary quantities, counts, age, mass, length, electrical current

ชนิดข้อมูลใน RapidMiner

- **Text** (ตัวหนังสือ) ข้อมูลถูกจัดเก็บเป็นข้อความ เป็นประโยค มีความหลากหลายของคำ
- **Nominal** ข้อมูลถูกตีความเป็นค่านาม
 - **Binominal** ข้อมูลถูกตีความเป็นค่านาม และแบ่งเป็นสองค่า เช่น yes/no, ชาย/หญิง
 - **Polynominal** ข้อมูลถูกตีความเป็น ค่านามเช่นเดียวกับ Binominal แต่มีมากกว่าสองค่าเช่น สี ดำ ชาย แดง, เล็ก กลาง ใหญ่
- **Numeric** (ตัวเลข) ถูกตีความเป็นตัวเลข
 - **Integer** ตัวเลขจำนวนเต็มเช่น 4-10
 - **Real** ตัวเลขจำนวนจริง เช่น 3.2 4.5 -5
- **Date_Time** ข้อมูลที่ถูกจัดเก็บหมายถึงวันที่ และเวลาโดยสามารถนำไปดำเนินการทางคณิตศาสตร์ เช่น 23.12.2014 17:59
 - **Date** หมายถึงวันที่จัดเก็บเป็น วันเดือนปี เช่น 23.12.2014
 - **Time** หมายถึงเวลาที่จัดเก็บเป็น ชั่วโมง นาที เช่น 17:59

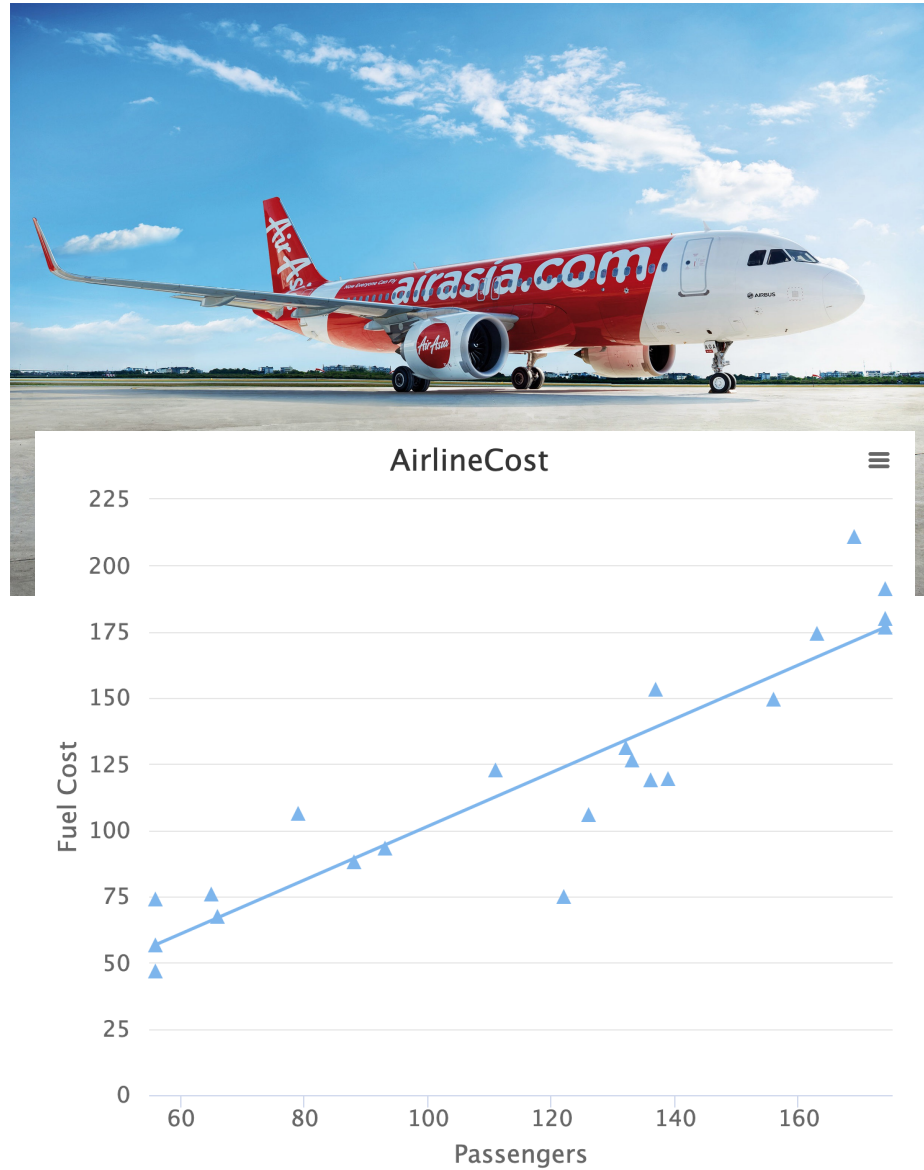
Regression

LAB 1

Linear Regression



Regression Model Airline Cost Data



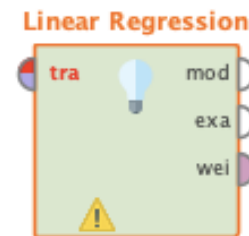
Linear Regression

Use Case: Airline Cost Dataset

- Create model Linear Regression



- exa = example
- ori = original

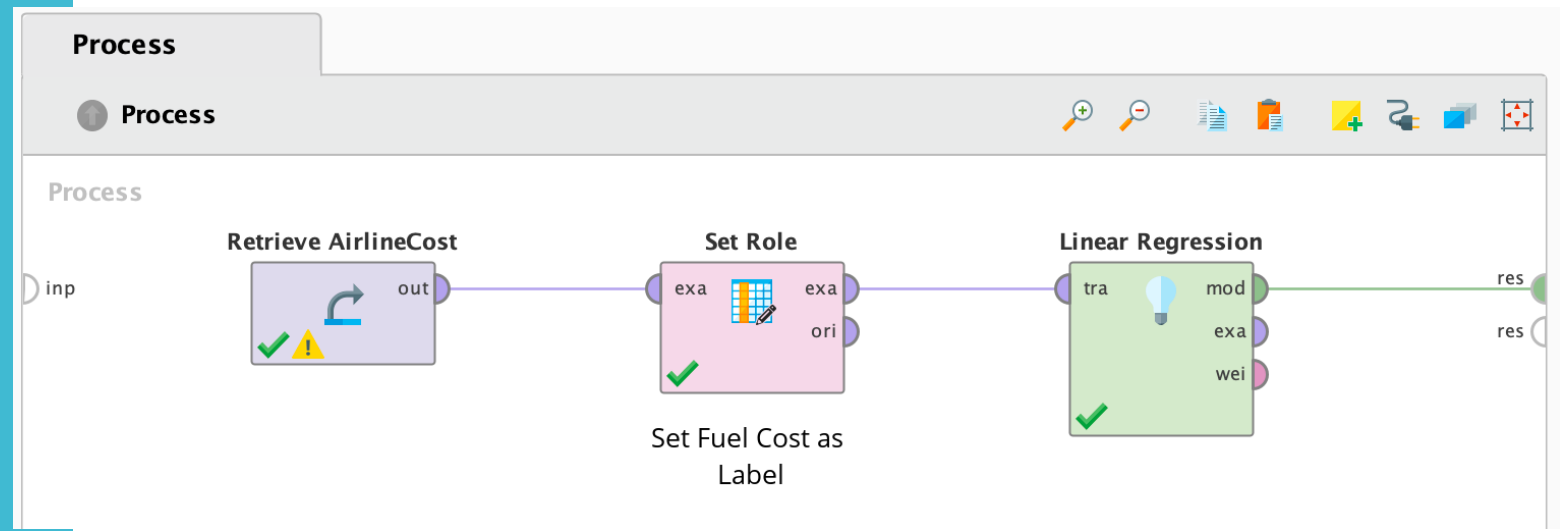


- tra = training set
- mod = model
- exa = exampleSet
- wei = weights

Linear Regression

Use Case: Airline Cost Dataset

- Create model Linear Regression

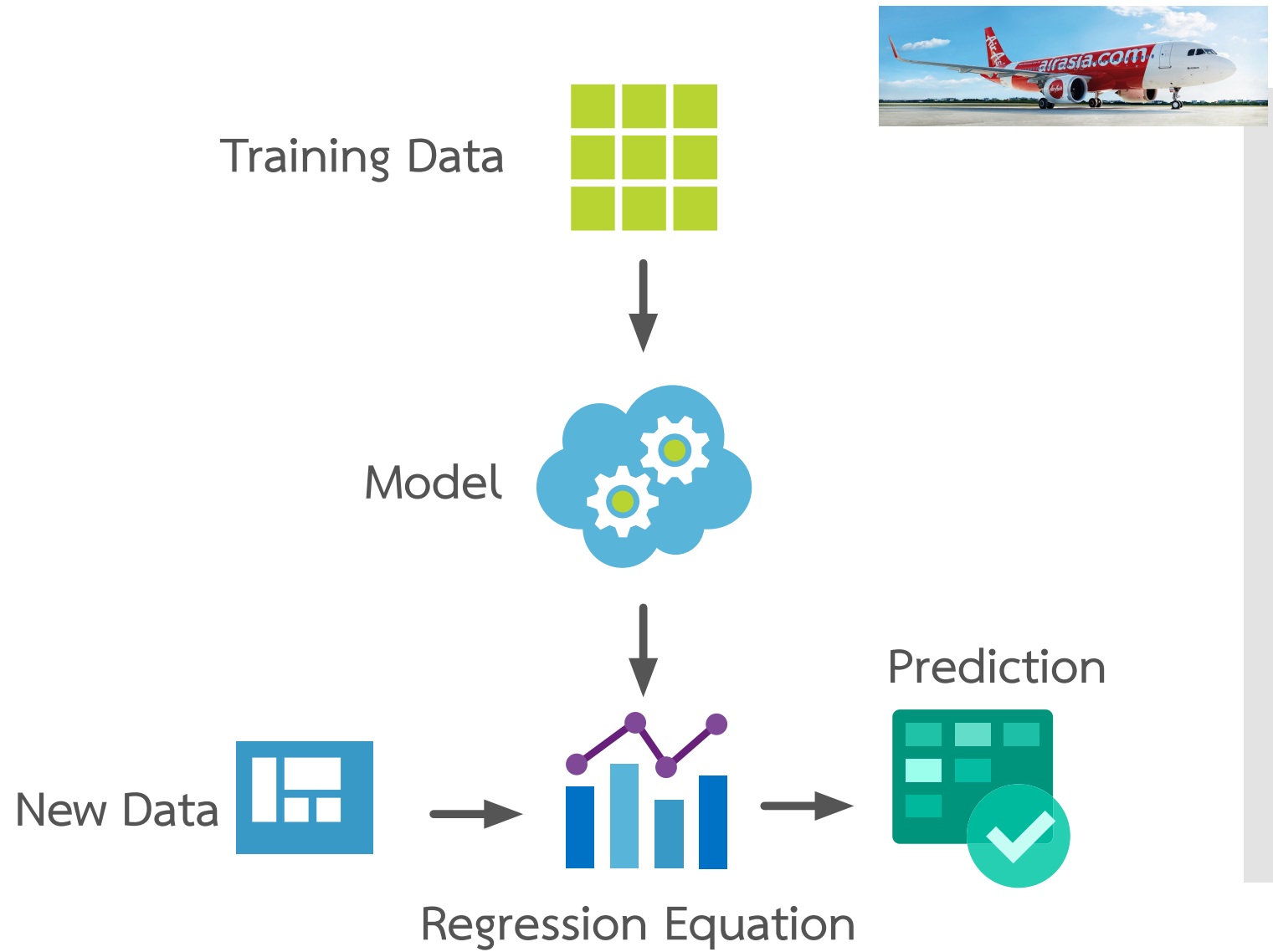


Linear Regression Model Result

Attribute	Coefficient	Std. Error
Passengers	1.015	0.115
(Intercept)	0.080	14.381

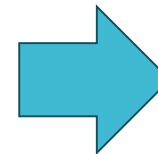
Regression Equation: $\hat{Y} = 0.08 + 1.015X$

Linear Regression Prediction



Create Test Data for Prediction

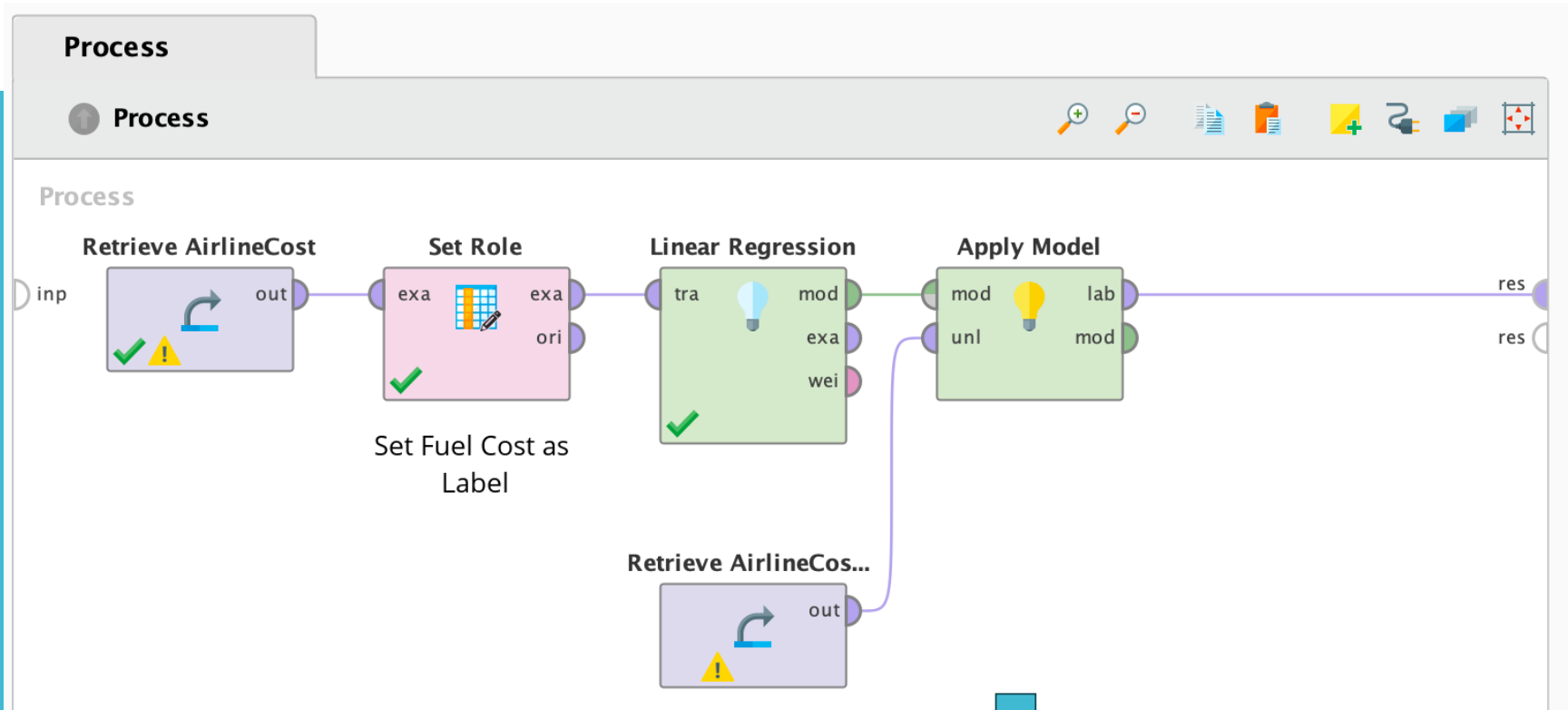
Data Editor		
Row No.	Passengers (integer) <i>regular</i>	Fuel Cost (real) <i>regular</i>
1	126	105.970
2	79	106.360
3	163	174.450
4	93	93.460
5	56	74.240
6	122	74.990
7	133	126.680
8	169	211.150
9	111	122.970
10	136	118.970
11	65	75.920
12	56	46.740
13	88	88.080
14	139	119.630
15	174	179.850
16	156	149.680
17	66	67.450
18	174	191.220
19	132	131.420
20	137	153.220



Data Editor		
Row No.	Passengers (integer) <i>regular</i>	
1	100	
2	500	

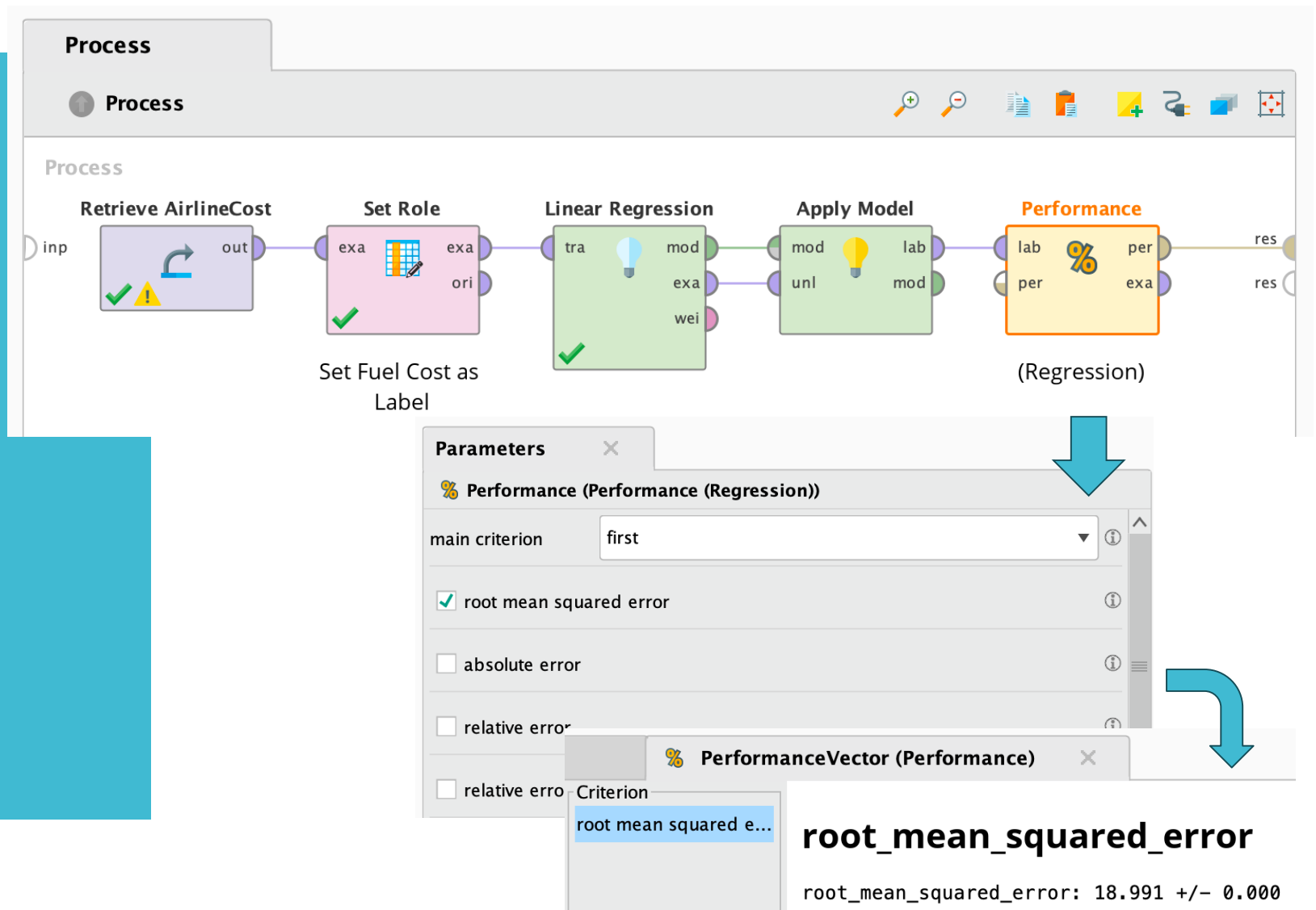


Apply model



Row No.	prediction(Fuel Cost)	Passengers
1	101.590	100
2	507.626	500

Performance Evaluation



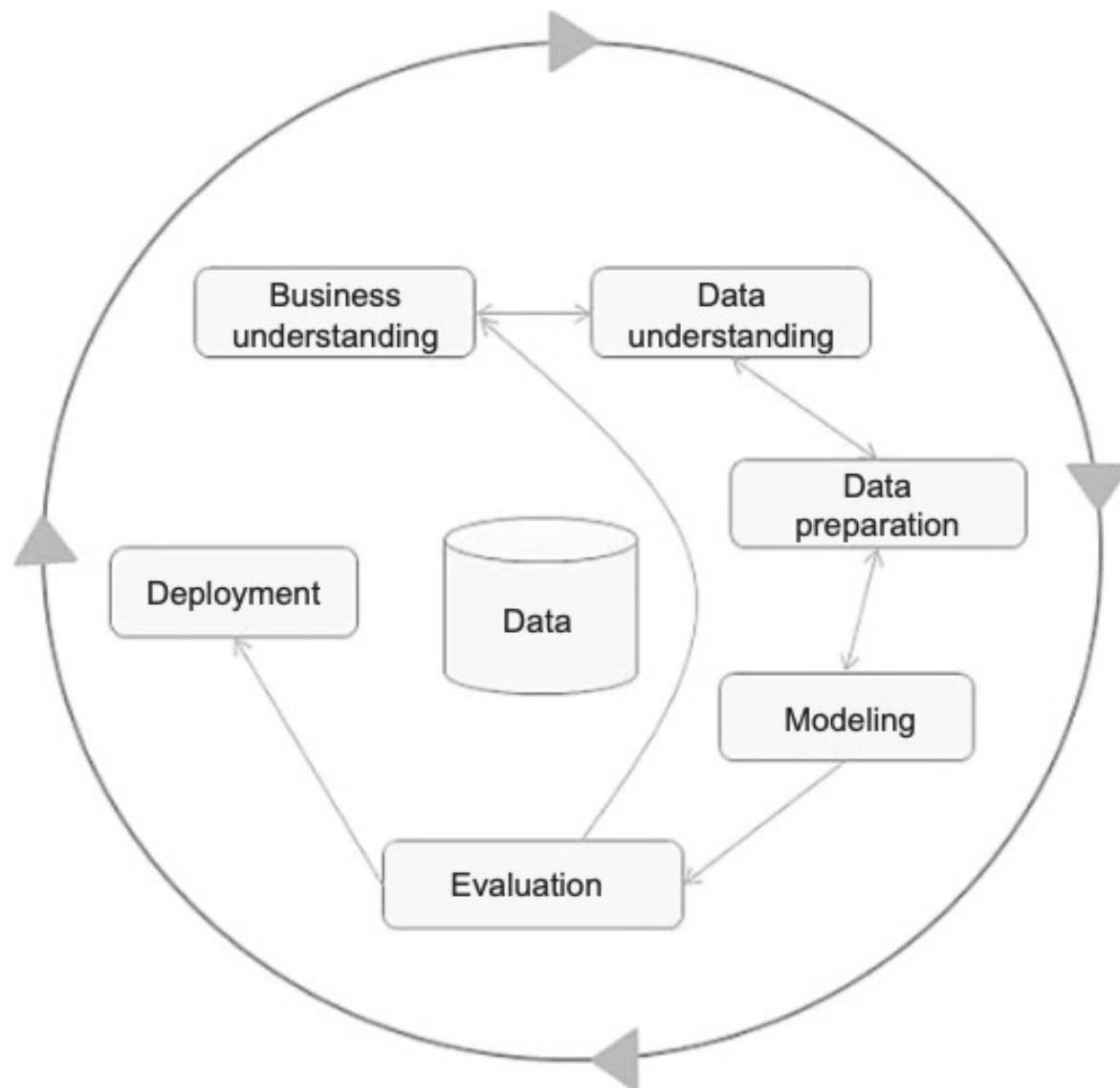
Regression

LAB 2

Multiple Linear Regression



กรอบงาน
Cross Industry
Standard Process
for Data Mining
(CRISP-DM)



1. Business

Understanding: Linear Regression

Use Case:

<https://www.kaggle.com/datasets/schirmerchad/bostonhousingmlnd?select=housing.csv>

- Source
- Harrison, D. and Rubinfeld, D.L. (1978) Hedonic prices and the demand for clean air. J. Environ. Economics and Management 5, 81–102.
- Belsley D.A., Kuh, E. and Welsch, R.E. (1980) Regression Diagnostics. Identifying Influential Data and Sources of Collinearity. New York: Wiley.

1.Business Understanding:

Full Dataset

Table 5.1 Sample View of the Classic Boston Housing Dataset

CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	Target = MEDV
0.00632	18	2.31	0	0.538	6.575	65.2	4.09	1	296	15.3	396.9	4.98	24
0.02731	0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	396.9	9.14	21.6
0.02729	0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	392.83	4.03	34.7
0.03237	0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63	2.94	33.4
0.06905	0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	396.9	5.33	36.2
0.02985	0	2.18	0	0.458	6.43	58.7	6.0622	3	222	18.7	394.12	5.21	28.7
0.08829	12.5	7.87	0	0.524	6.012	66.6	5.5605	5	311	15.2	395.6	12.43	22.9
0.14455	12.5	7.87	0	0.524	6.172	96.1	5.9505	5	311	15.2	396.9	19.15	27.1

1.Business

Understanding:

Table 5.2 Attributes of Boston Housing Dataset

1. CRIM per capita crime rate by town
2. ZN proportion of residential land zoned for lots over 25,000 sq.ft.
3. INDUS proportion of non-retail business acres per town
4. CHAS Charles River dummy variable (=1 if tract bounds river; 0 otherwise)
5. NOX nitric oxide concentrations (parts per 10 million)
6. RM average number of rooms per dwelling
7. AGE proportion of owner-occupied units built prior to 1940
8. DIS weighted distances to five Boston employment centers
9. RAD index of accessibility to radial highways
10. TAX full-value property-tax rate per \$10,000
11. PTRATIO pupil-teacher ratio by town
12. $B = 1000(B_k - 0.63)^2$ where B_k is the proportion of blacks by town
13. LSTAT percent lower status of the population
14. MEDV Median value of owner-occupied homes in \$1000's

2.Data

Understanding:
Import data

Import Data - Specify your data format

Specify your data format

☒ Header Row

Start Row

Column Separator

File Encoding

Escape Character

Decimal Character

☒ Use Quotes

☐ Trim Lines

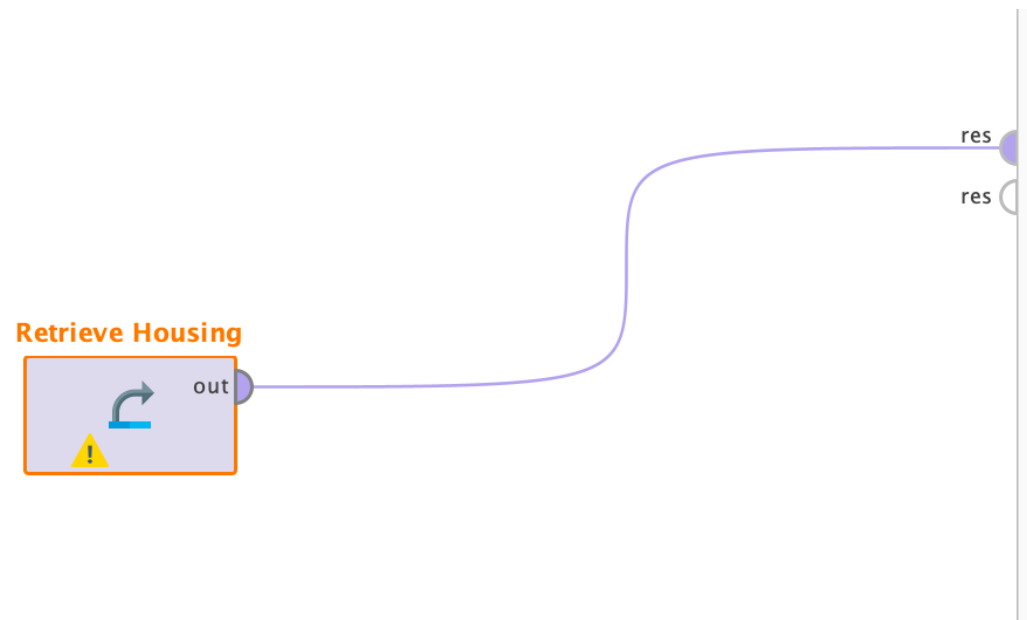
☒ Skip Comments

	RM	LSTAT	PTRATIO	MEDV
1				
2	6.575	4.98	15.3	504000.0
3	6.421	9.14	17.8	453600.0
4	7.185	4.03	17.8	728700.0
5	6.998	2.94	18.7	701400.0
6	7.147	5.33	18.7	760200.0
7	6.43	5.21	18.7	602700.0
8	6.012	12.43	15.2	480900.0
9	6.172	19.15	15.2	569100.0
10	5.631	29.93	15.2	346500.0
11	6.004	17.1	15.2	396900.0
12	6.377	20.45	15.2	315000.0

✓ no problems.

2.Data

Understanding: Import data



- <https://docs.rapidminer.com/latest/studio/operators/>

2.Data Understanding: Data Import

The screenshot displays the RapidMiner Studio interface. At the top, a toolbar contains several icons, with a blue play button icon highlighted by a red rectangle. Below the toolbar, the 'Views' section includes tabs for 'Design', 'Results', 'Turbo Prep', 'Auto Model', and 'Deployments'. The 'Design' tab is active, showing a 'Process' canvas with a 'Retrieve housing' operator connected to the input and output ports. The 'Repository' panel on the left shows a tree structure with 'Local Repository (Local)' expanded, revealing 'data' and 'processes' folders. The 'Operators' panel on the left lists various functions, with 'Linear Regression' highlighted. The 'Parameters' panel on the right shows settings for the 'Process' operator, including 'logverbosity', 'logfile', 'resultfile', and 'random seed'. The 'Help' panel on the right provides a synopsis and description of the 'Process' operator.

Repository

- Import Data
- Time Series
- Tutorials
- Community Samples (connected)
- Local Repository (Local)
 - Connections
 - data
 - housing (6/8/22 1:10 AM - 17 kB)
 - processes
- Temporary Repository (Local)
- DB (Legacy)

Operators

Search for Operators

- Functions (9)
 - Generalized Linear Model
 - Linear Regression
 - Polynomial Regression
 - Function Fitting
 - Vector Linear Regression
 - Local Polynomial Regression
 - Seemingly Unrelated Regression
 - Gaussian Process

[Get more operators from the Marketplace](#)

Process

Process

Retrieve housing

inp out res res

Parameters

Process

- logverbosity: init
- logfile:
- resultfile:
- random seed: 2001

[Hide advanced parameters](#)

[Change compatibility \(9.10.001\)](#)

Help

Process

RapidMiner Studio Core

Synopsis

The root operator which is the outer most operator of every process.

Description

Each process must contain exactly one operator of this class, and it must be the root operator of the process. This operator provides a set of parameters that are of global relevance to the process like logging and initialization parameters of the random number generator.

Recommended Operators

- Set Role 35%
- Select Attrib... 35%
- Apply Model 26%

<new process> – RapidMiner Studio Educational 9.10.001 @ Macs-MacBook-Air.local

Views: Design Results Turbo Prep Auto Model Deployments

Find data, operators...etc All Studio ▾

Result History ExampleSet (//Local Repository/data/housing) ✕

Open in Turbo Prep Auto Model

Filter (489 / 489 examples): all ▾

Row No.	RM	LSTAT	PTRATIO	MEDV
1	6.575	4.980	15.300	504000
2	6.421	9.140	17.800	453600
3	7.185	4.030	17.800	728700
4	6.998	2.940	18.700	701400
5	7.147	5.330	18.700	760200
6	6.430	5.210	18.700	602700
7	6.012	12.430	15.200	480900
8	6.172	19.150	15.200	569100
9	5.631	29.930	15.200	346500
10	6.004	17.100	15.200	396900
11	6.377	20.450	15.200	315000
12	6.009	13.270	15.200	396900
13	5.889	15.710	15.200	455700
14	5.949	8.260	21	428400
15	6.096	10.260	21	382200
16	5.834	8.470	21	417900

ExampleSet (489 examples, 0 special attributes, 4 regular attributes)

Table 5.2 Attributes of Boston Housing Dataset

1. CRIM per capita crime rate by town
2. ZN proportion of residential land zoned for lots over 25,000 sq.ft.
3. INDUS proportion of non-retail business acres per town
4. CHAS Charles River dummy variable (= 1 if tract bounds river; 0 otherwise)
5. NOX nitric oxide concentrations (parts per 10 million)
6. RM average number of rooms per dwelling
7. AGE proportion of owner-occupied units built prior to 1940
8. DIS weighted distances to five Boston employment centers
9. RAD index of accessibility to radial highways
10. TAX full-value property-tax rate per \$10,000
11. PTRATIO pupil-teacher ratio by town
12. $B 1000(B_k - 0.63)^2$ where B_k is the proportion of blacks by town
13. LSTAT percent lower status of the population
14. MEDV Median value of owner-occupied homes in \$1000's

2.Data

Understanding: Descriptive Statistics

<new process> – RapidMiner Studio Educational 9.10.001 @ Macs-MacBook-Air.local

Views: Design Results Turbo Prep Auto Model Deployments Find data, operators...etc All Studio

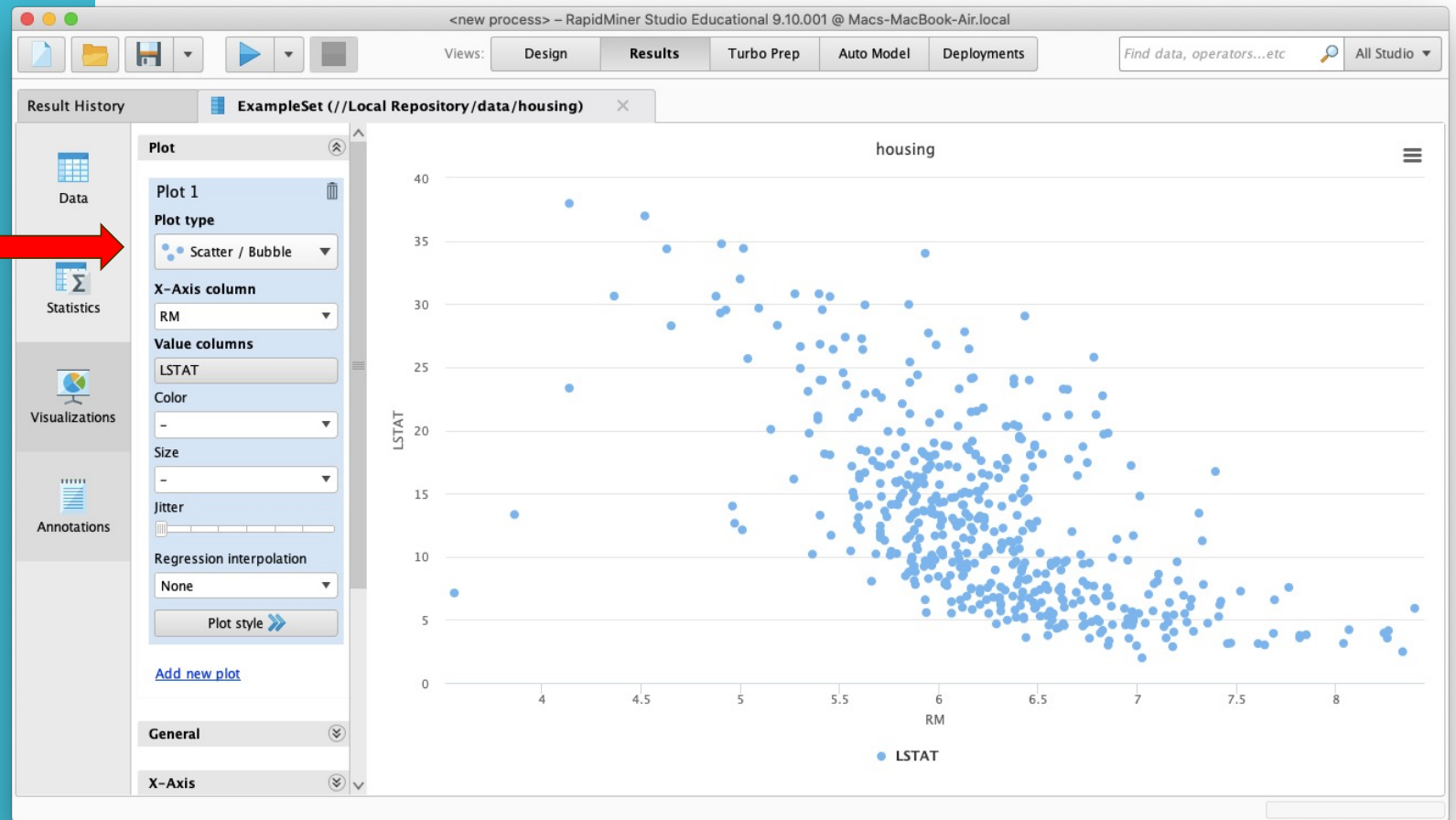
Result History ExampleSet (//Local Repository/data/housing) X

Filter (4 / 4 attributes): Search for Attribute.

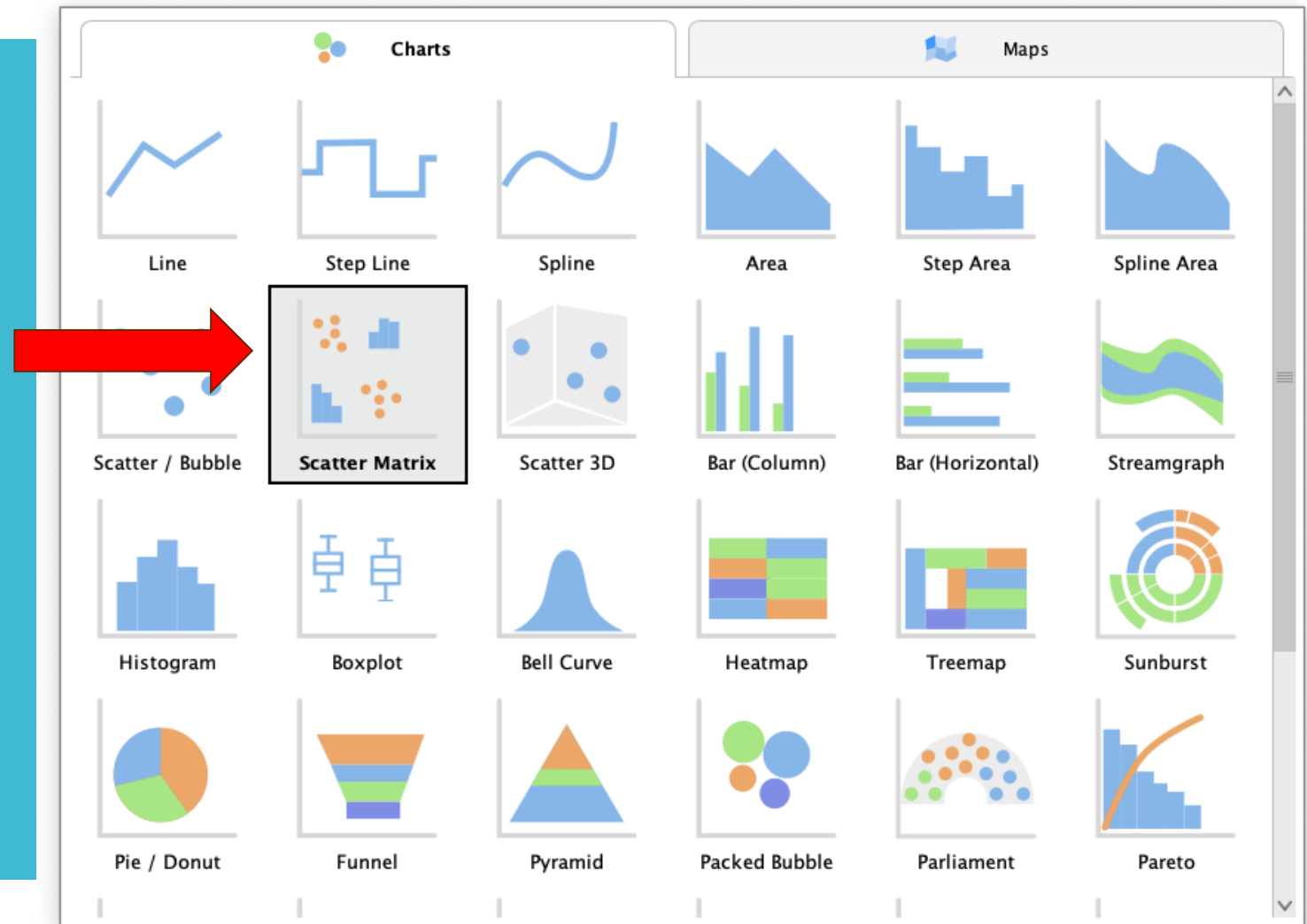
Name	Type	Missing	Statistics		
RM	Real	0	Min 3.561	Max 8.398	Average 6.240
LSTAT	Real	0	Min 1.980	Max 37.970	Average 12.940
PTRATIO	Real	0	Min 12.600	Max 22	Average 18.517
MEDV	Real	0	Min 105000	Max 1024800	Average 454342.945

Showing attributes 1 - 4 Examples: 489 Special Attributes: 0 Regular Attributes: 4

2.Data Understanding: Visualization

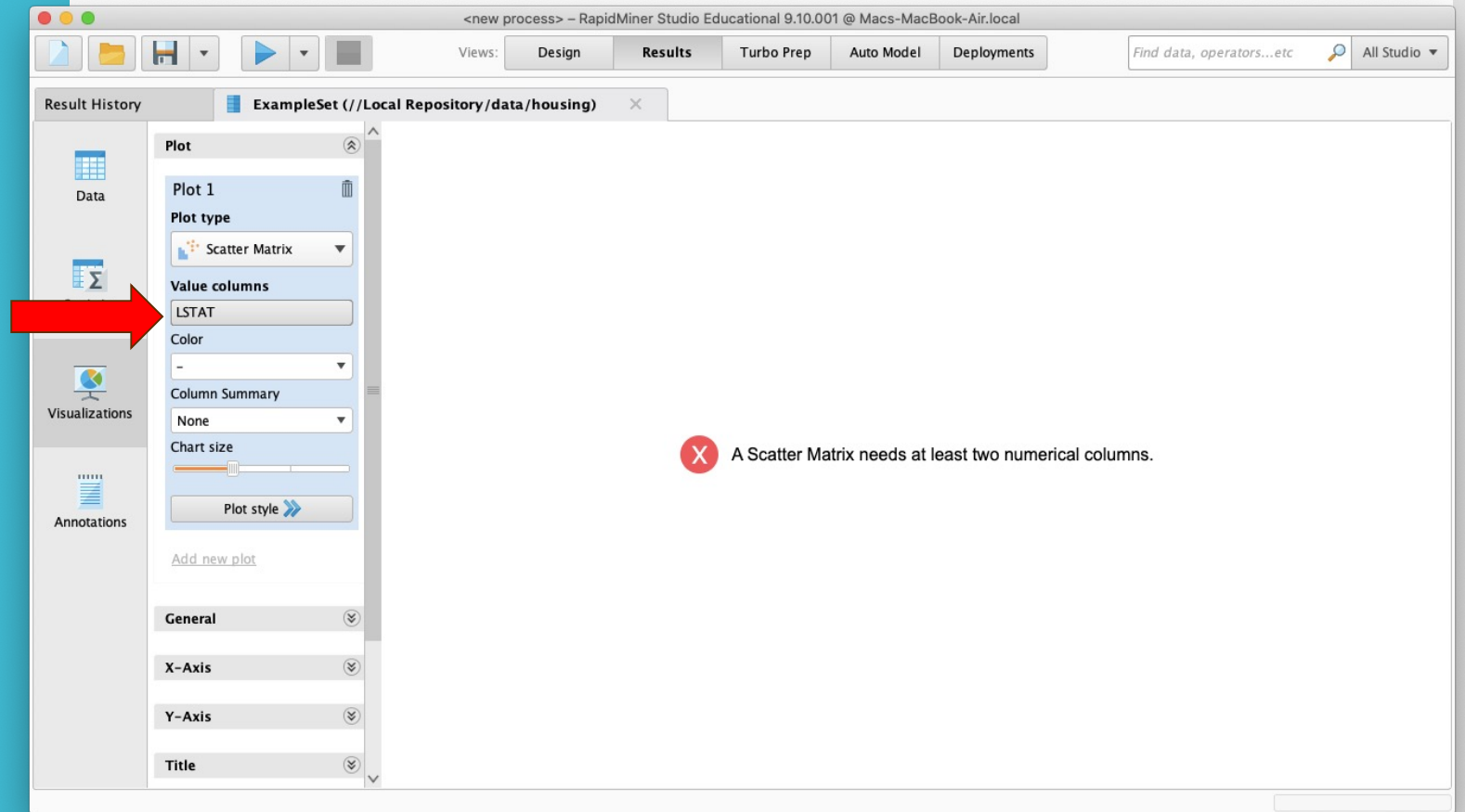


2.Data Understanding: Visualization

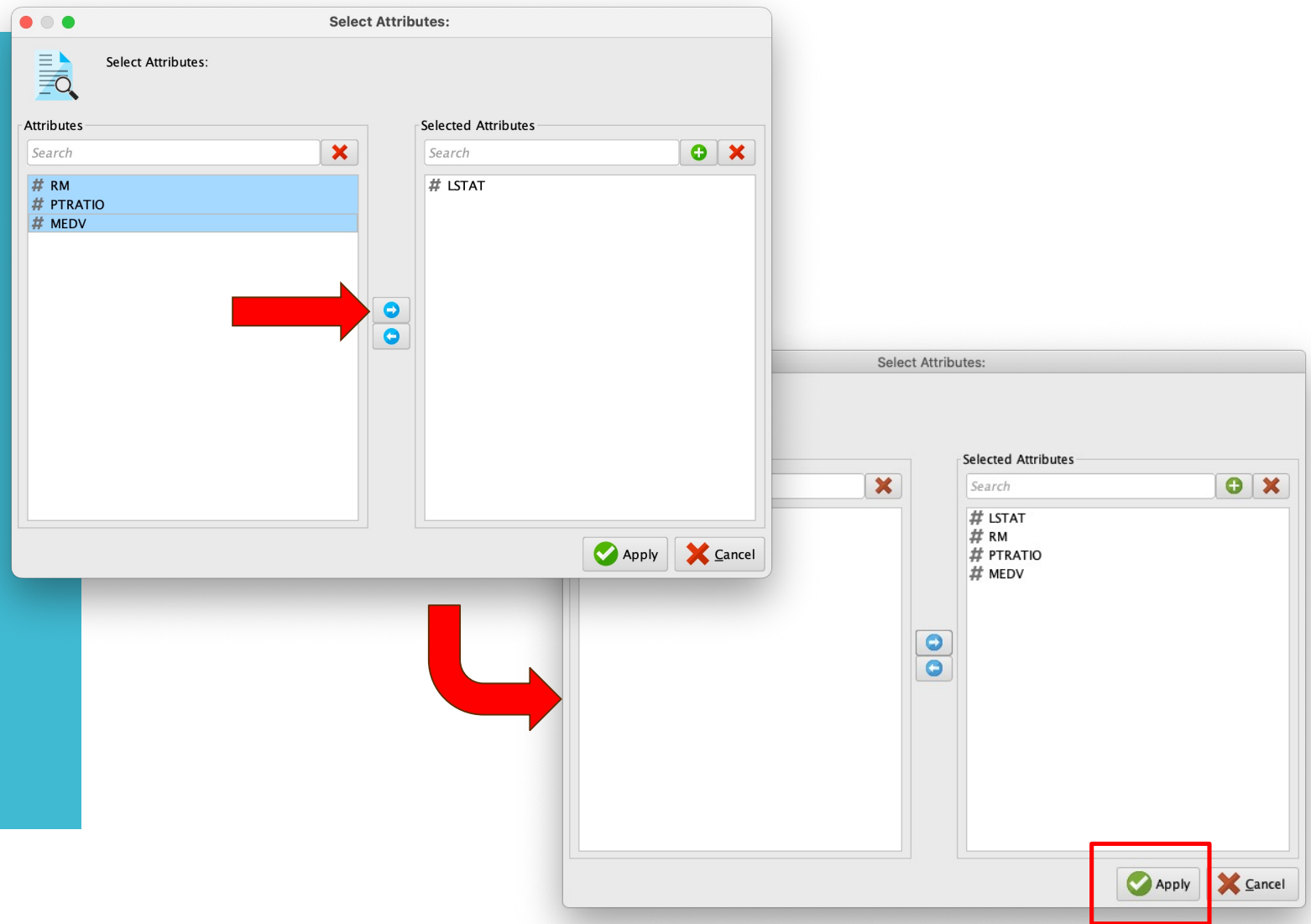


2.Data

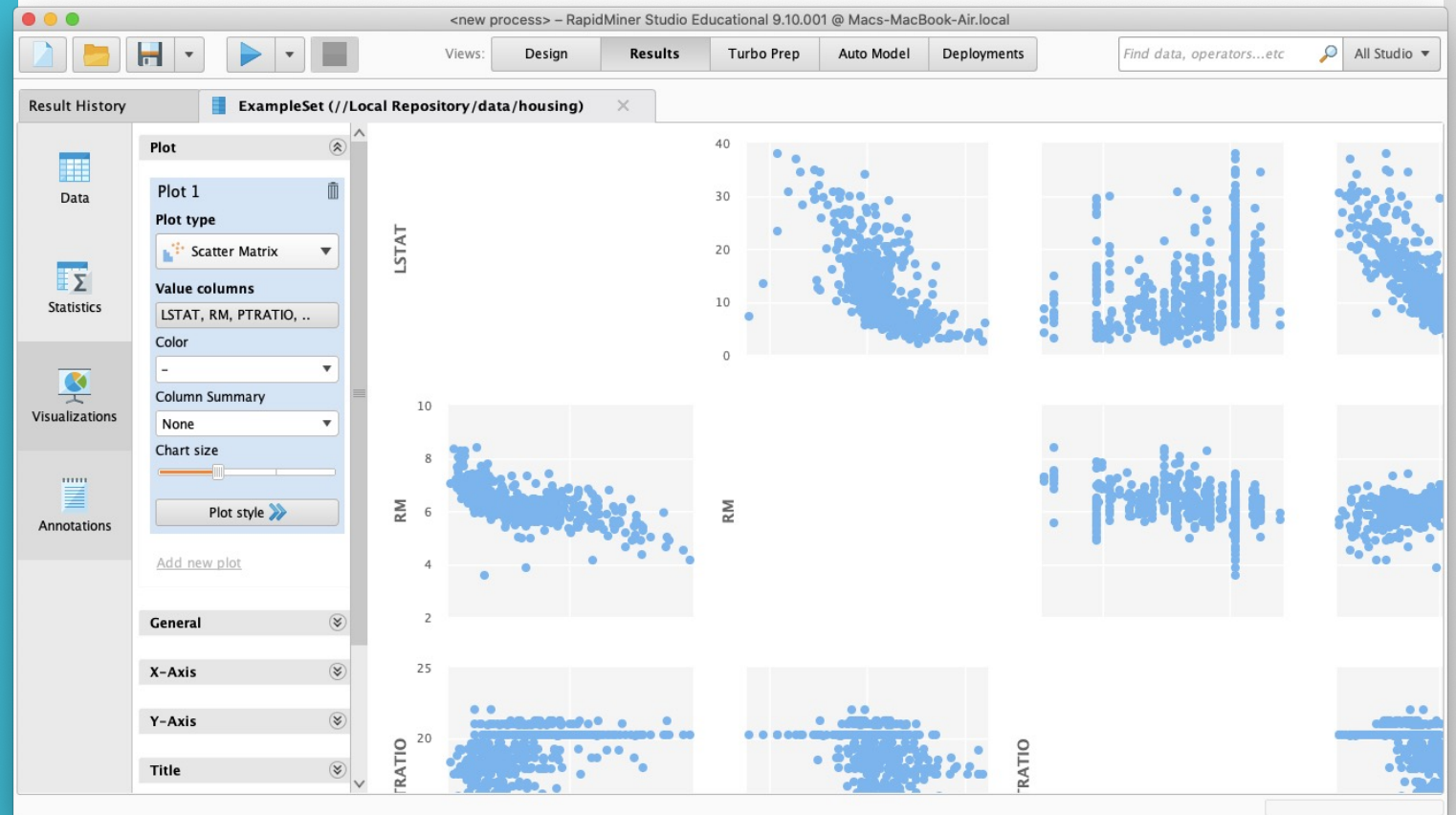
Understanding: Visualization



2.Data Understanding: Visualization



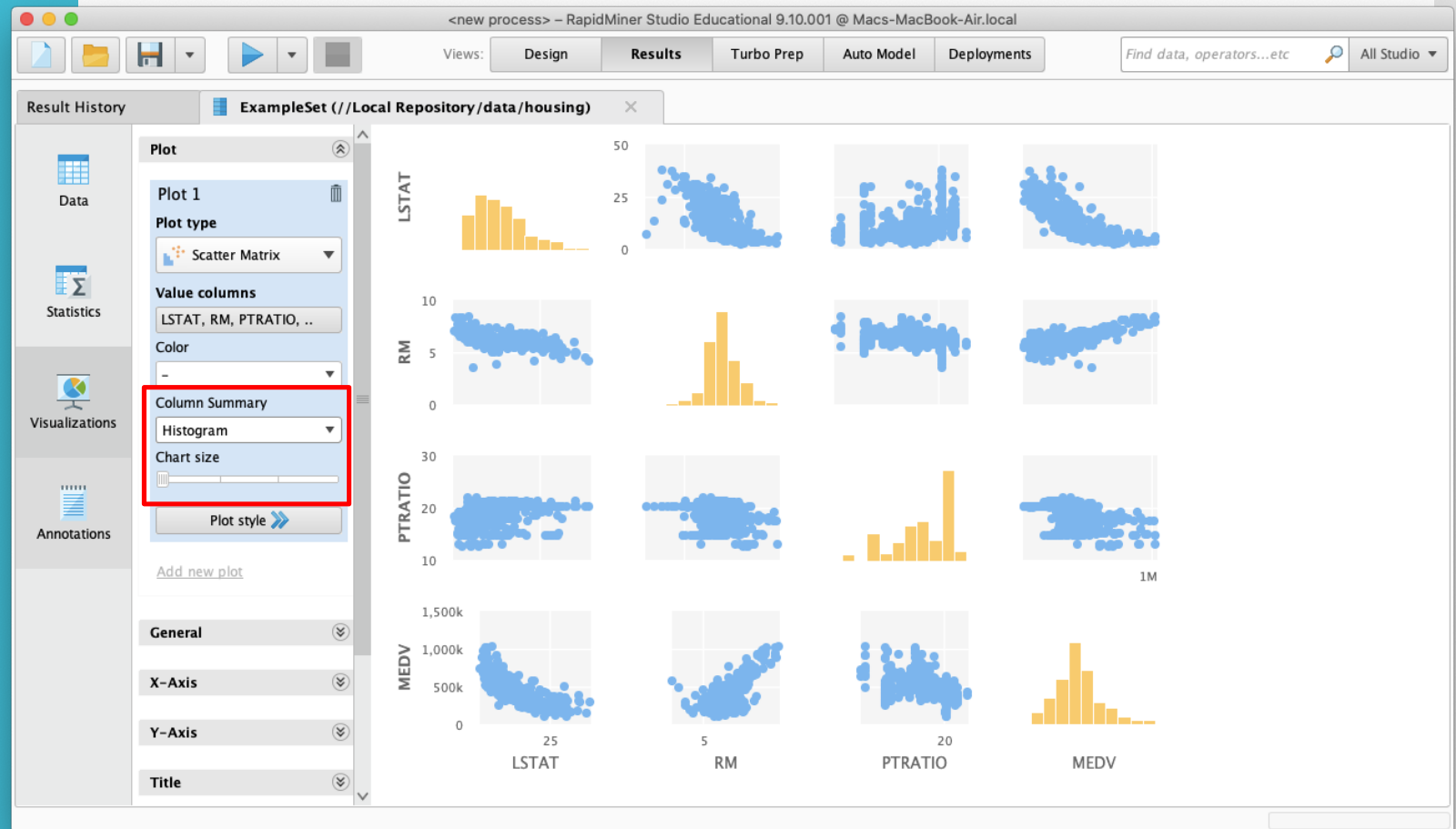
2.Data Understanding: Visualization



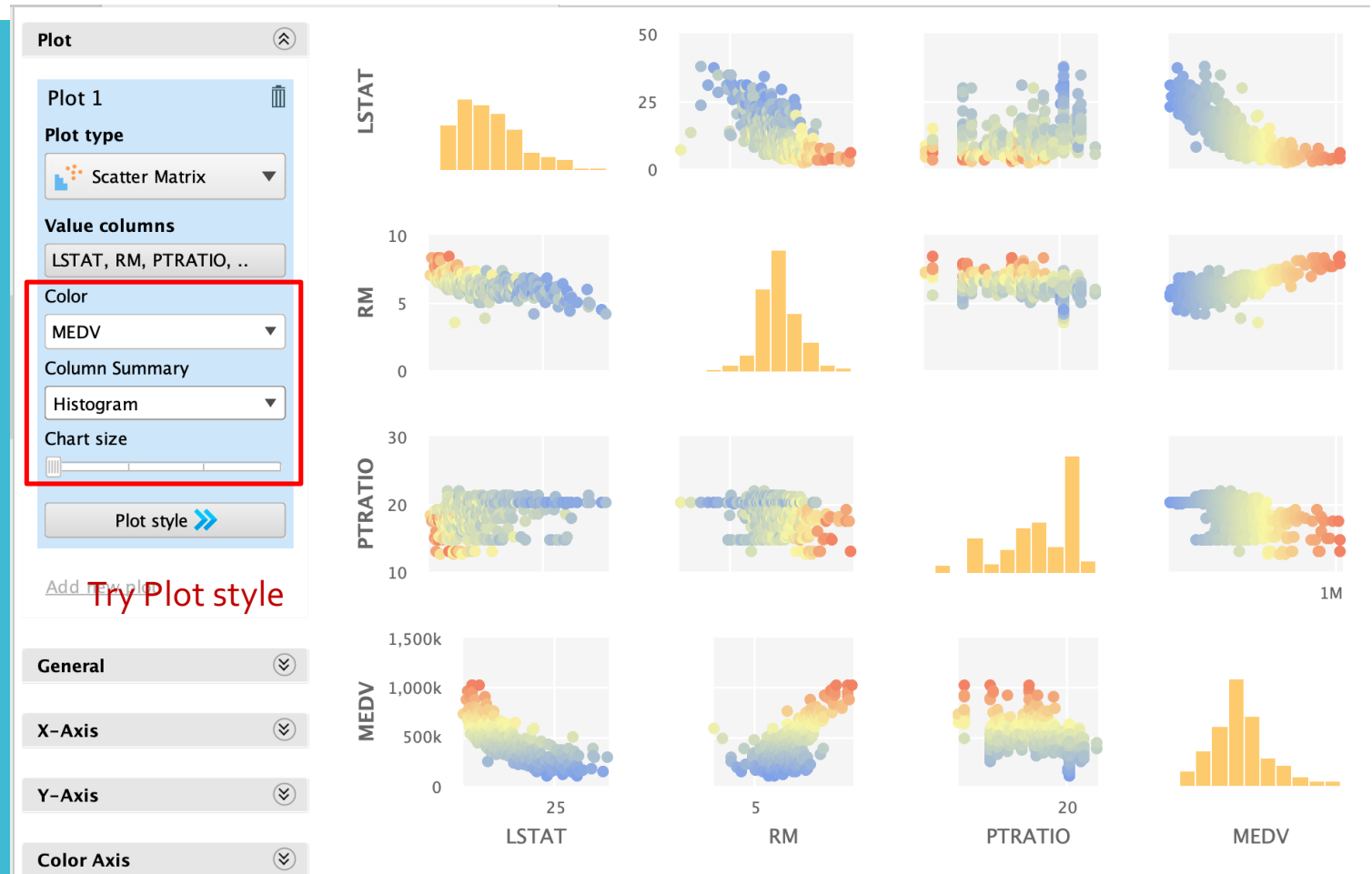
2.Data Understanding: Visualization



2.Data Understanding: Visualization

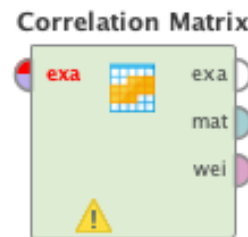


2.Data Understanding: Visualization



2.Data

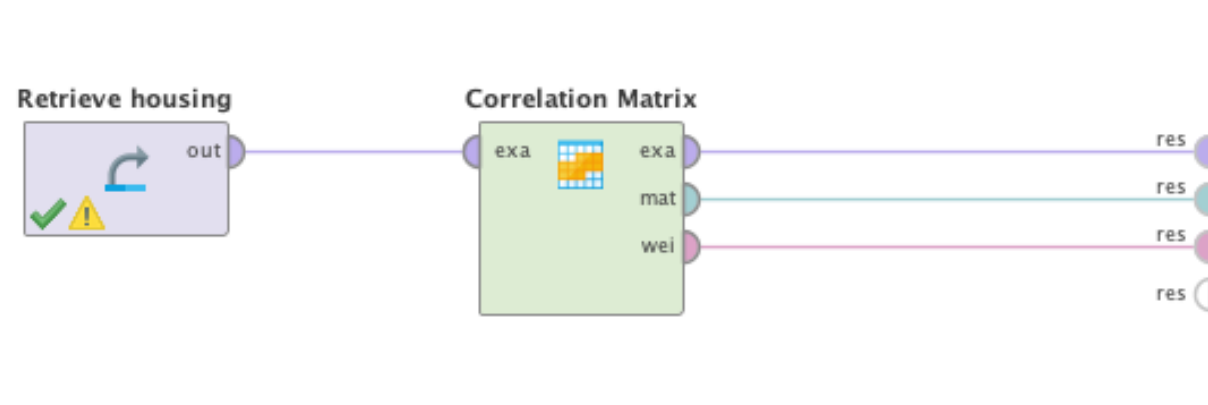
Understanding: Correlation Matrix



- exa = example set
- mat = matrix
- wei = weight

2.Data

Understanding: Correlation Matrix



Correlation Matrix (Correlation Matrix) ExampleSet (Retrieve housing)

Result History AttributeWeights (Correlation Matrix)

Data

Pairwise Table

Matrix Visualization

Annotations

Attributes	RM	LSTAT	PTRATIO	MEDV
RM	1	-0.612	-0.305	0.697
LSTAT	-0.612	1	0.360	-0.761
PTRATIO	-0.305	0.360	1	-0.519
MEDV	0.697	-0.761	-0.519	1

3.Data Preparation: Set Role



- exa = example
- ori = original

3.Data Preparation: Set Role

The screenshot displays the RapidMiner Studio Educational 9.10.001 interface. The main workspace shows a process diagram with two operators: 'Retrieve housing' and 'Set Role'. The 'Set Role' operator is highlighted, and its parameters are shown in the right-hand pane. The parameters are: attribute name (MEDV), target role (label), and set additional roles (Edit List (0)...). The 'Help' pane on the right provides information about the 'Set Role' operator, including its tags, synopsis, and description.

Repository

- Import Data
- Training Resources (connected)
- Samples
- Community Samples (connected)
- Local Repository (Local)
 - Connections
 - data
 - boston (6/8/22 4:43 PM - 61 kB)
 - housing (6/8/22 1:10 AM - 17 kB)
 - housing (6/8/22 10:17 AM - 861 bytes)
 - processes

Operators

set role

- Blending (1)
 - Attributes (1)
 - Names & Roles (1)
 - Set Role

Process

Process

Retrieve housing

Set Role

Drag here

Parameters

Set Role

attribute name: MEDV

target role: label

set additional roles: Edit List (0)...

Change compatibility (9.10.001)

Help

Set Role

RapidMiner Studio Core

Tags: Label, Target, Id, Class, Dependent, Independent, Special, Regular, Inputs, Columns, Attributes, Features, Variables, Types, Names & Roles

Synopsis

This Operator is used to change the role of one or more Attributes.

[Jump to Tutorial Process](#)

Description

The role of an Attribute describes how other Operators handle this Attribute. The default role is regular, other roles are classified as

Recommended Operators

- Select At... 46%
- Apply M... 32%
- Filter Ex... 26%

3.Data Preparation: Set Role

<new process*> - RapidMiner Studio Educational 9.10.001 @ Macs-MacBook-Air.local

Views: Design Results Turbo Prep Auto Model Deployments Find data, operators...etc All Studio

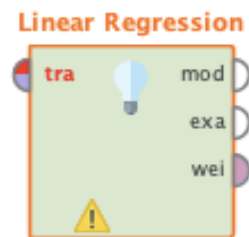
Result History ExampleSet (Set Role) ExampleSet (//Local Repository/data/housing)

Open in Turbo Prep Auto Model Filter (489 / 489 examples): all

Row No.	MEDV	RM	LSTAT	PTRATIO
1	504000	6.575	4.980	15.300
2	453600	6.421	9.140	17.800
3	728700	7.185	4.030	17.800
4	701400	6.998	2.940	18.700
5	760200	7.147	5.330	18.700
6	602700	6.430	5.210	18.700
7	480900	6.012	12.430	15.200
8	569100	6.172	19.150	15.200
9	346500	5.631	29.930	15.200
10	396900	6.004	17.100	15.200
11	315000	6.377	20.450	15.200
12	396900	6.009	13.270	15.200
13	455700	5.889	15.710	15.200
14	428400	5.949	8.260	21
15	382200	6.096	10.260	21
16	417900	5.834	8.470	21

ExampleSet (489 examples, 1 special attribute, 3 regular attributes)

4.Modeling: Linear Regression



- tra = training set
- mod = model
- exa = exampleSet
- wei = weights

4.Modeling: Linear Regression

The screenshot displays the RapidMiner Studio Educational 9.10.001 interface. The main workspace shows a workflow with three operators: 'Retrieve housing', 'Set Role', and 'Linear Regression'. The 'Linear Regression' operator is selected, and its parameters are visible on the right. The 'Repository' panel on the left shows the 'housing' dataset loaded. The 'Operators' panel at the bottom left shows the 'Linear Regression' operator selected under the 'Modeling' category. The 'Help' panel on the right provides a synopsis and description of the 'Linear Regression' operator.

Repository

- Training Resources (connected)
- Samples
- Community Samples (connected)
- Local Repository (Local)
 - Connections
 - data
 - boston (6/8/22 4:43 PM - 61 kB)
 - housing (6/8/22 1:10 AM - 17 kB)
 - housing (6/8/22 10:17 AM - 861 bytes)
 - processes

Process

Process

Retrieve housing → Set Role → Linear Regression

Parameters

- Linear Regression
 - feature selection: none
 - ☒ eliminate colinear features
 - min tolerance: 0.05
 - ☒ use bias
 - ridge: 1.0E-8
 - [Hide advanced parameters](#)

Help

Linear Regression
RapidMiner Studio Core

Tags: Supervised, Classification, Regression, Model, Least squares, Ordinary, Ridge, Ols, Glim, Generalized, Functions

Synopsis
This operator calculates a linear regression model from the input ExampleSet.
[Jump to Tutorial Process](#)

Description
Regression is a technique used for numerical prediction. Regression is a statistical measure that attempts to determine the

Recommended Operators

- Apply M... 52%
- Perform... 51%
- Select At... 47%

4.Modeling: Linear Regression

<new process*> - RapidMiner Studio Educational 9.10.001 @ Macs-MacBook-Air.local

Views: Design Results Turbo Prep Auto Model Deployments Find data, operators...etc All Studio

Result History LinearRegression (Linear Regression) ExampleSet (//Local Repository/data/housing)

Attribute	Coefficient	Std. Error	Std. Coefficient	Tolerance	t-Stat	p-Value	Code
RM	86565.236	7888.895	0.337	0.630	10.973	0	****
LSTAT	-10849.340	732.138	-0.465	0.600	-14.819	0	****
PTRATIO	-19492.116	2039.047	-0.249	0.860	-9.559	0	****
(Intercept)	415464.397	68845.713	?	?	6.035	0.000	****

4. Modeling: Linear Regression

The screenshot displays the RapidMiner Studio Educational 9.10.001 interface. The window title is "<new process*> - RapidMiner Studio Educational 9.10.001 @ Macs-MacBook-Air.local". The top toolbar includes icons for file operations and execution. The "Views" tab is set to "Results". The "Result History" pane on the left shows the "LinearRegression (Linear Regression)" process selected. The main area displays the results for this process, including a "Data" section with a table of values and a "Description" section with a list of features.

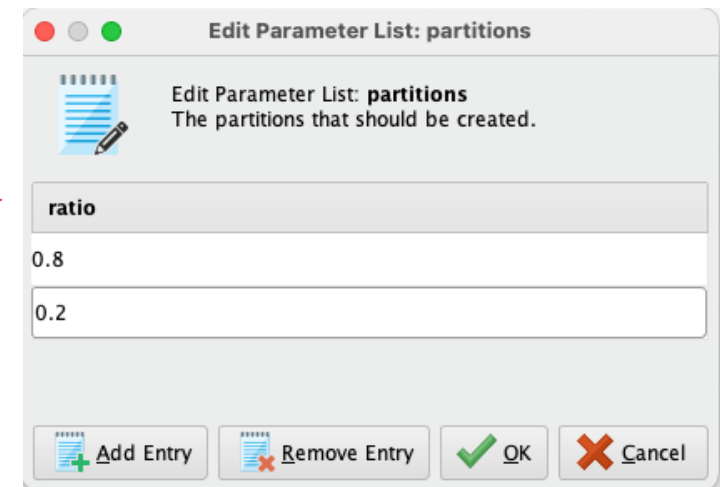
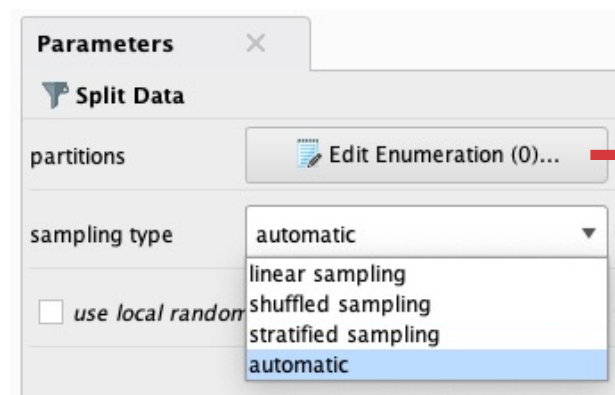
LinearRegression

86565.236	*	RM
- 10849.340	*	LSTAT
- 19492.116	*	PTRATIO
+ 415464.397		

Description

Annotations

4. Modeling: Linear Regression Performance



4.Modeling:

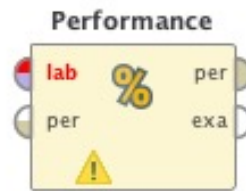
Apply Model



- mod = model In/Out
- unl = unlabelled data
- lab = labelled data

4.Modeling:

Performance (Regression)

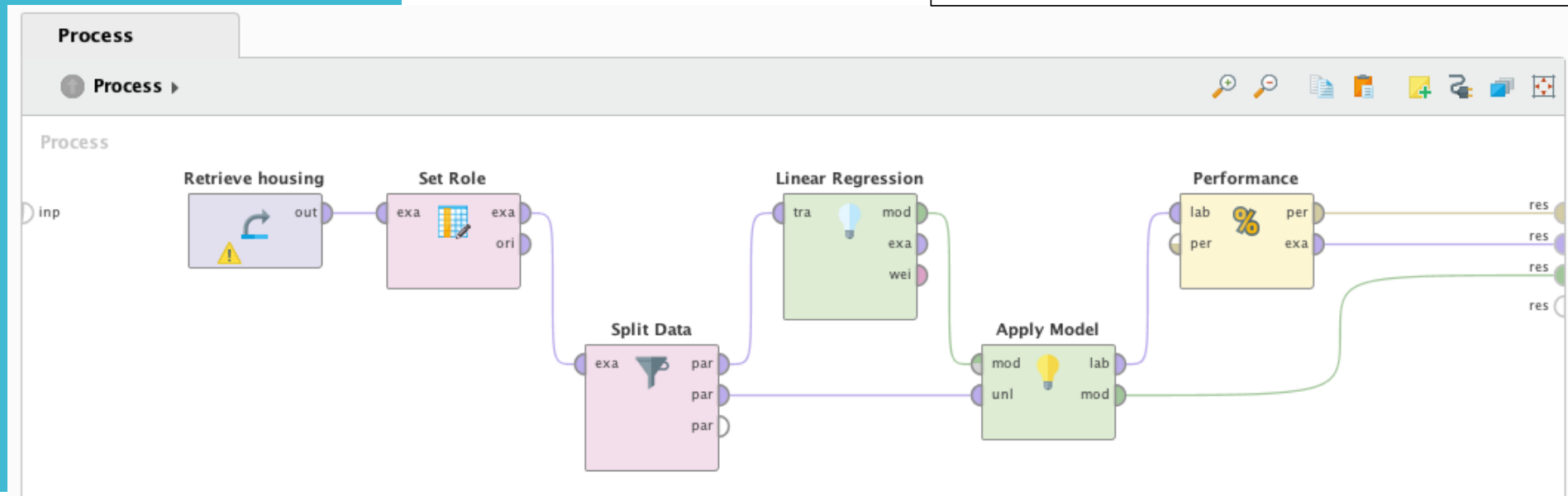


- lab = labelled data
- per = performance
- exa = example set

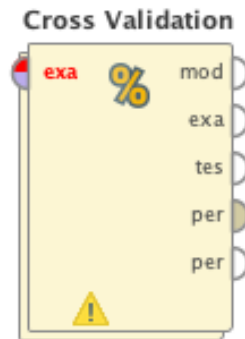
4.Modeling: Linear Regression Performance

PerformanceVector

PerformanceVector:
root_mean_squared_error: 98253.253 +/- 0.000
absolute_error: 74343.061 +/- 64240.260
relative_error: 18.48% +/- 19.00%
squared_error: 9653701814.249 +/- 16223783926.451
correlation: 0.819
squared_correlation: 0.672



5.Evaluation: Validation



mod	= model
exa	= example set
tes	= test result set
per	= performance

Differentiation

- **Split Validation**

This Operator is similar to the Cross Validation Operator but only splits the data into one training and one test set. Hence it is similar to one iteration of the cross validation.

See page 945 for details.

- **Split Data**

This Operator splits an ExampleSet into different subsets. It can be used to manual perform a validation.

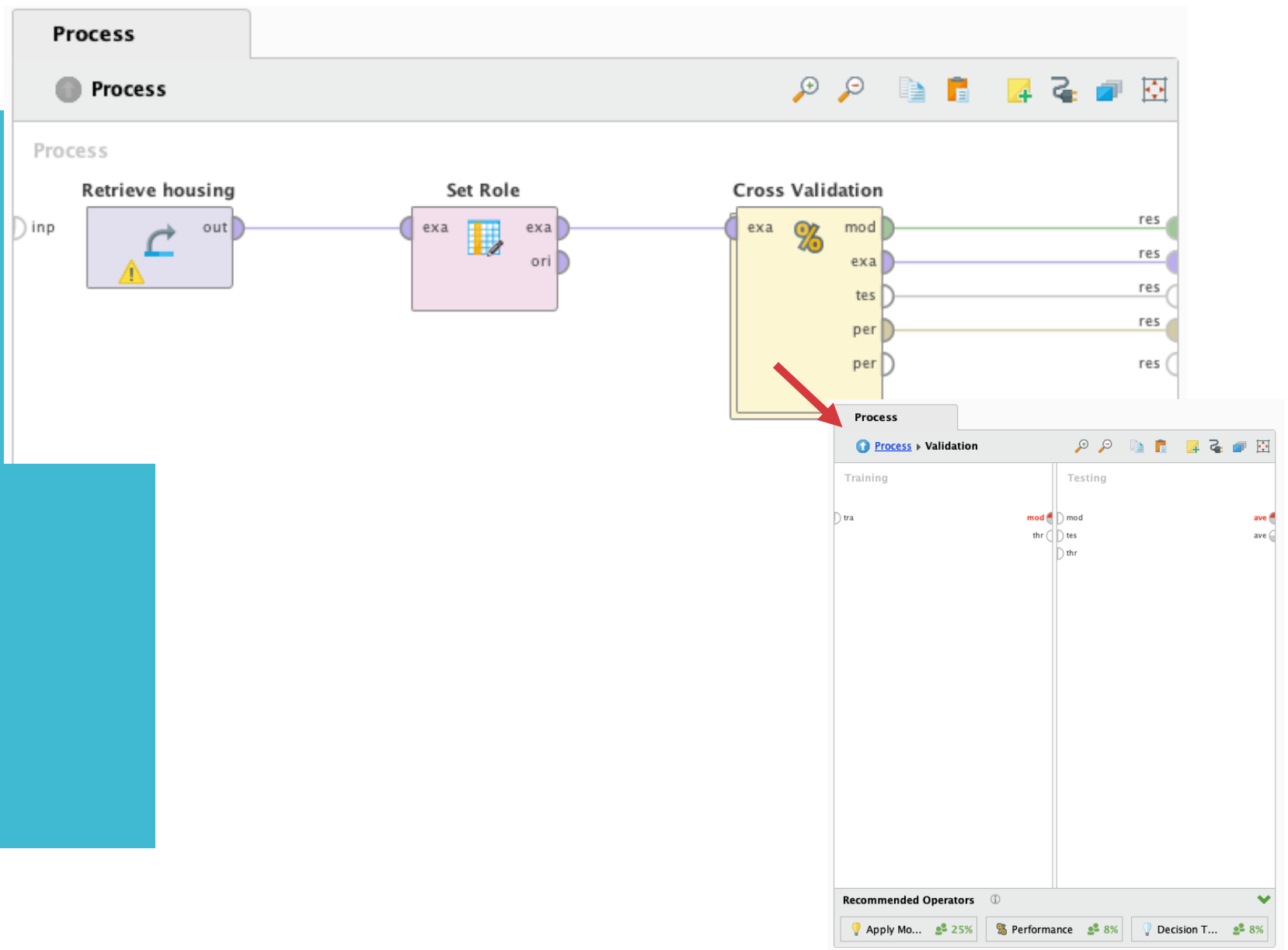
See page 285 for details.

- **Bootstrapping Validation**

This Operator is similar to the Cross Validation Operator. Instead of splitting the input ExampleSet into different subset, the Bootstrapping Validation Operator uses bootstrapping sampling to get the training data. Bootstrapping sampling is sampling with replacement.

See page 937 for details.

5.Evaluation: Validation



5.Evaluation: Validation

The screenshot displays the RapidMiner Studio Educational 9.10.001 interface on a Mac. The main workspace shows a process design for validation, divided into Training and Testing phases. The Training phase includes a 'Linear Regression' operator, and the Testing phase includes an 'Apply Model' operator followed by a 'Performance' operator. The 'Performance' operator is highlighted with a red box. The left sidebar contains the 'Repository' and 'Operators' panels. The 'Repository' panel shows the 'housing' dataset selected. The 'Operators' panel shows the 'Performance (Regression)' operator selected. The right sidebar contains the 'Parameters' and 'Help' panels. The 'Parameters' panel shows the 'main criterion' set to 'first' and several performance metrics checked: 'root mean squared error', 'absolute error', and 'relative error'. The 'Help' panel provides a synopsis of the 'Performance (Regression)' operator, stating it is used for statistical performance evaluation of regression tasks and delivers a list of performance criteria values.

Repository

- Time Series
- Tutorials
- Community Samples (connected)
- Local Repository (Local)
 - Connections
 - data
 - housing (6/8/22 1:10 AM - 17 kB)
 - processes
- Temporary Repository (Local)
- DB (Legacy)

Operators

performance

- validation (20)
 - Performance (18)
 - Predictive (7)
 - Performance (Classification)
 - Performance (Binominal Classification)
 - Performance (Regression)
 - Performance (Costs)
 - Performance (Ranking)
 - Performance (Support Vector Count)

We found "Model Management" in the Marketplace. [Show me!](#)

Process

Views: Design Results Turbo Prep Auto Model Deployments

Find data, operators...etc All Studio

Training

tra → Linear Regression (mod, tra, wei, exa) → mod → thr → tes → thr

Testing

tes → thr → Apply Model (mod, tes, unl, lab, mod) → lab → per → ave → exa → ave

Parameters

Performance (Performance (Regression))

main criterion: first

- ☒ root mean squared error
- ☒ absolute error ✓
- ☒ relative error ✓
- ☐ relative error lenient

[Hide advanced parameters](#)

Help

Performance (Regression)

RapidMiner Studio Core

Tags: RMSE, Errors, Absolute, Relative, Squared, Predictive

Synopsis

This operator is used for statistical performance evaluation of regression task and delivers a list of performance criteria values of the regression task.

[Jump to Tutorial Process](#)

Recommended Operators

- Performance (Cl... 31%
- Performance 25%
- Performance (Bi... 23%

5.Evaluation: Validation

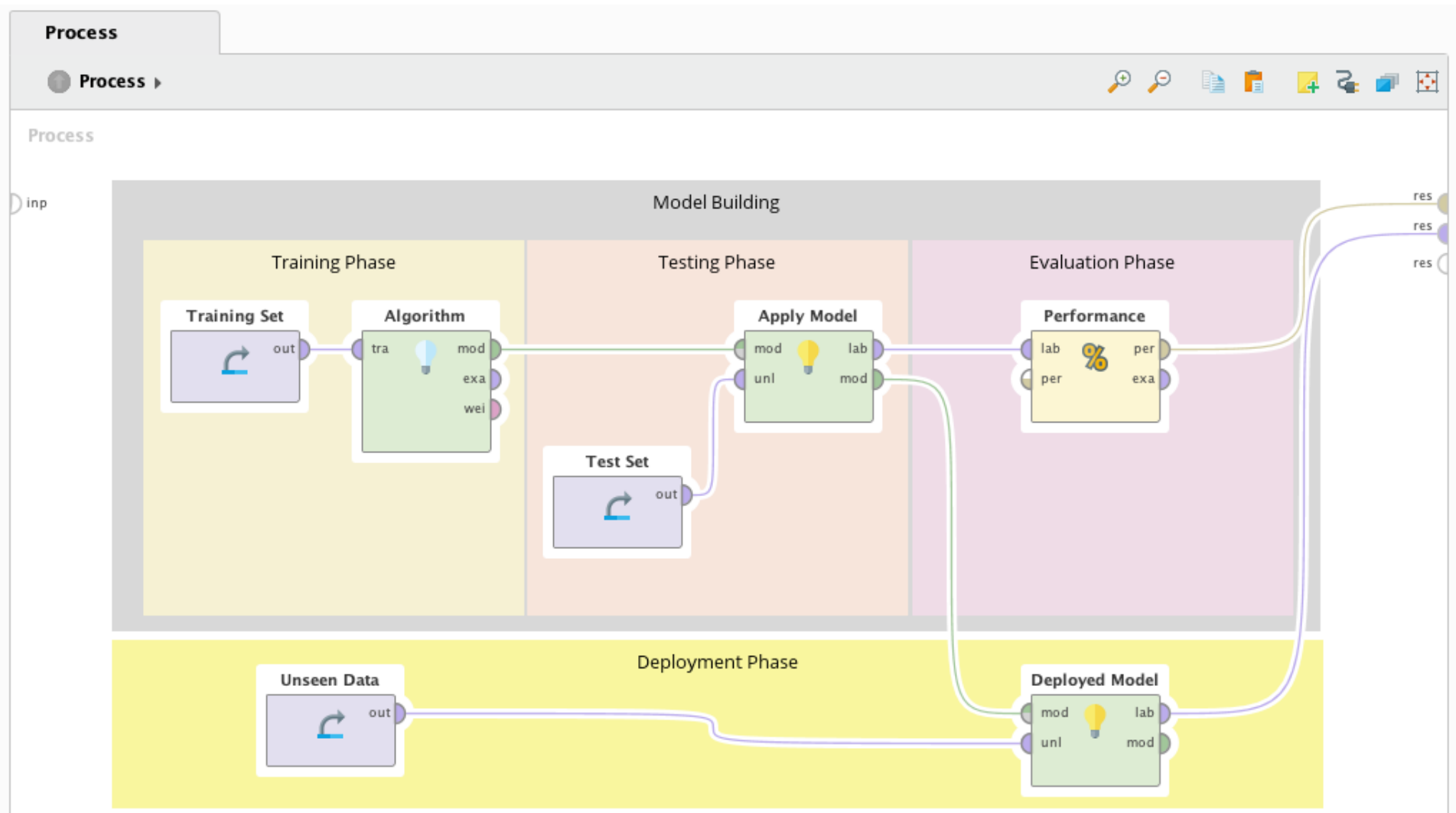
The screenshot shows the Orange3 software interface. The top toolbar includes icons for file operations and a 'Views' dropdown menu. The 'Results' tab is active, showing a list of views: 'LinearRegression (Linear Regression)', 'ExampleSet (//Local Repository/data/housing)', 'PerformanceVector (Performance)', and 'ExampleSet (Set Role)'. The 'PerformanceVector (Performance)' view is selected, displaying the 'root_mean_squared_error' criterion. The left sidebar shows the 'Performance' view selected, with a list of criteria: 'root mean squared e...', 'absolute error', 'relative error', 'squared error', 'correlation', and 'squared correlation'. The main area displays the 'root_mean_squared_error' value: 87766.956 +/- 15241.703 (micro average: 88959.133 +/- 0.000).

The screenshot shows the 'PerformanceVector' view in the Orange3 software. The left sidebar shows the 'Performance' view selected, with a list of criteria: 'Performance', 'Description', and 'Annotations'. The main area displays the 'PerformanceVector' results:

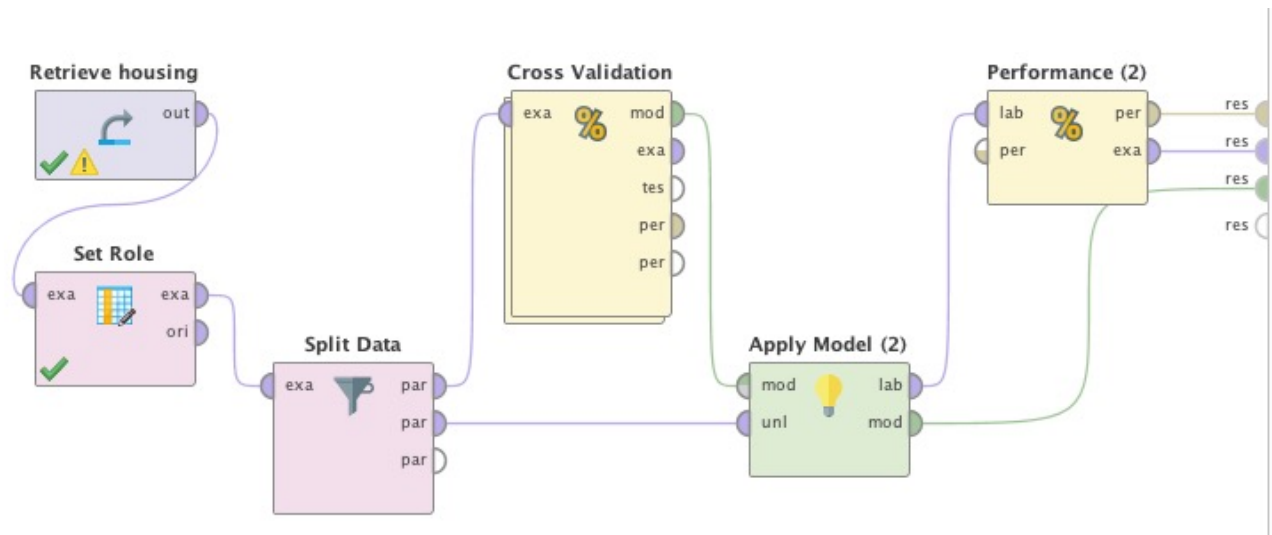
PerformanceVector

PerformanceVector:
root_mean_squared_error: 87766.956 +/- 15241.703 (micro average: 88959.133 +/- 0.000)
absolute_error: 66077.131 +/- 11219.963 (micro average: 66076.253 +/- 59562.203)
relative_error: 17.42% +/- 3.33% (micro average: 17.42% +/- 20.51%)
squared_error: 7912117057.660 +/- 2725418526.365 (micro average: 7913727278.736 +/- 14952568404.240)
correlation: 0.847 +/- 0.046 (micro average: 0.843)
squared_correlation: 0.719 +/- 0.075 (micro average: 0.710)

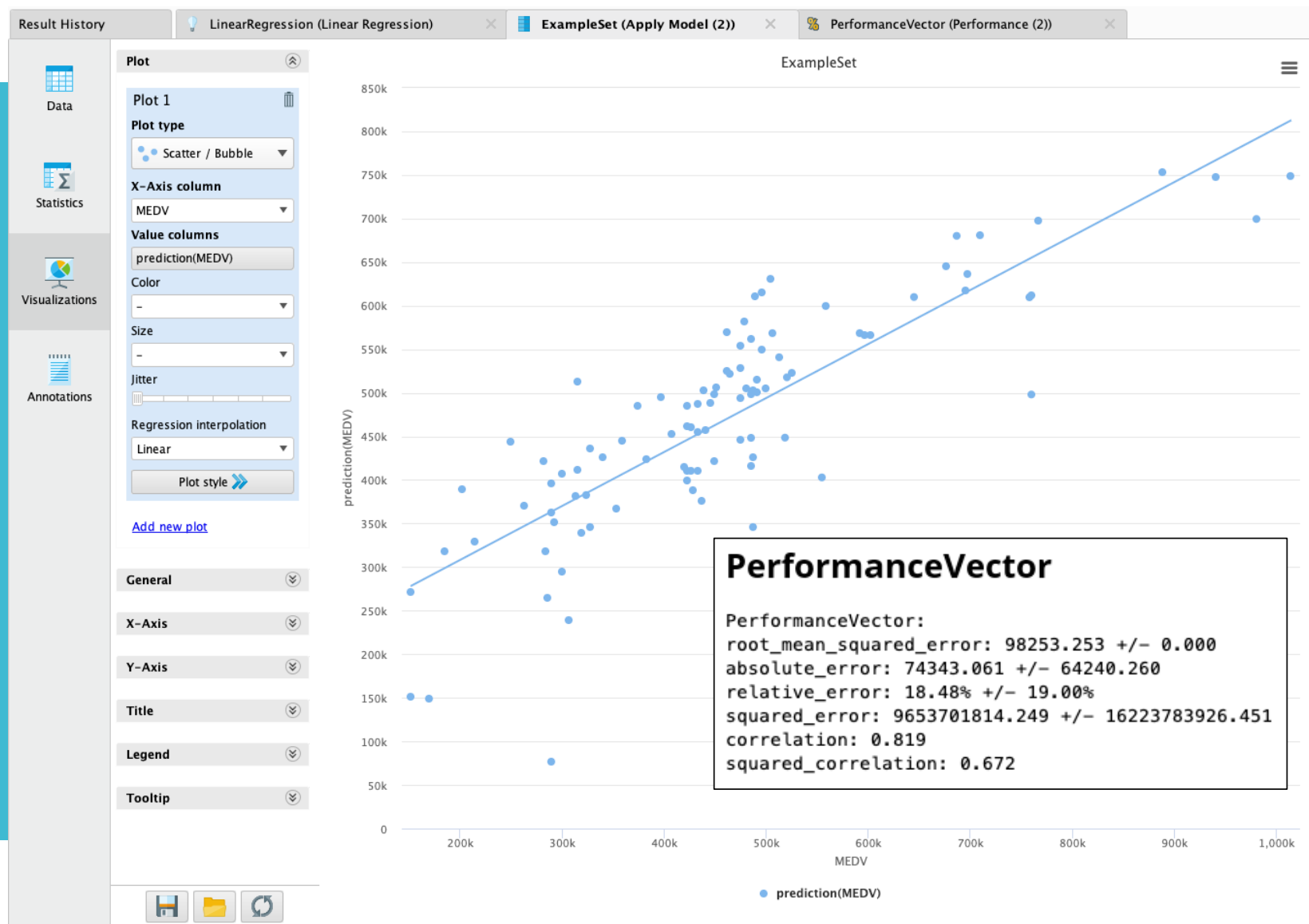
6. Deployment: Workflow



6. Deployment: Prediction



6. Deployment: Prediction



การทดลอง



- ทดลองทำนายค่า **MEDV** จาก Dataset: boston.csv โดยดำเนินการวิเคราะห์ตามขั้นตอน CRISP-DM ด้วย Software Rapidminer AI Studio
- Capture ผลการทดลองแต่ละขั้นตอนทำเป็นรายงานส่งในรูปแบบ pdf